



DS 341
Course Project
Inventory Optimisation and Customer Segmentation

Members

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2. Dataset Overview

Name: Online Retail Dataset

Source: UCI Machine Learning Repository

Link: <https://archive.ics.uci.edu/dataset/352/online+retail>

File: Online Retail.xlsx

Records: 541,909 transactions

Features: 8 columns

(InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country)

Time Period: December 2010 - December 2011

Domain: E-commerce (UK-based online retail)

Problem Statement

Analyze customer purchasing behavior in an e-commerce dataset to:

1. Identify product co-purchase patterns using association rule mining
2. Segment customers based on their purchase behavior (RFM analysis)
3. Build predictive models to classify high-value customers
4. Provide actionable business recommendations to improve revenue and customer retention

Business Objectives

- Understand what products are frequently bought together
- Identify distinct customer segments for targeted marketing
- Predict which customers are likely to be high-value spenders
- Optimize inventory and marketing strategies

3. Preprocessing Details

Data preprocessing is crucial for ensuring data quality and preparing the dataset for analysis. This section covers:

Handling missing values

Feature engineering

Removing cancelled transactions

Filtering out invalid entries

Since the dataset that was chosen had no missing values, we did not have to implement any handling for missing values.

Key Preprocessing Steps

1. Remove rows with missing CustomerID (essential for customer segmentation)
2. Filter out cancelled orders (InvoiceNo starting with 'C')
3. Remove junk/system entries (POST, BANK CHARGES, etc.)
4. Validate data quality (positive quantities and prices only)

Feature Encoding and Transformation

Feature transformation is applied within the "Customer Segmentation" section to ensure the K-Means clustering algorithm functions correctly. Specifically, two distinct transformations are chained together on the RFM (Recency, Frequency, Monetary) data. First, a **Log Transformation** (`np.log(rfm + 1)`) is applied to address the "skewness" inherent in financial data, where a small number of customers often account for a large portion of the revenue (Pareto distribution). This compresses the wide range of values—making the data distribution more Gaussian (bell-shaped)—which is a key assumption for K-Means performance. Following this, **Standardization** (`StandardScaler`) is applied to the log-transformed data. This scales the features so they share a common mean of 0 and standard deviation of 1. Without this step, the "Monetary" feature (often in the thousands) would vastly overpower smaller features like "Frequency" during the distance calculations, leading to biased and inaccurate clusters.

Feature encoding in this project is implemented through custom logic rather than standard library encoders, specifically to tailor the data for Association Rule Mining and Classification. To implement Association Rule Mining, a **Binarization** process is used to create a market basket matrix: the code applies a custom function `encode_units` that converts product quantities into a simple binary format (1 if bought, 0 if not). This effectively acts as a One-Hot encoding scheme required by the Apriori algorithm, which needs to know *presence* rather than *quantity*. Additionally, in the "Classification" section, **Target Label Encoding** is performed to create the variable `IsHighValue`. The code converts the continuous "Monetary" variable into a categorical binary target (0 or 1) based on a percentile threshold. This manual encoding transforms a regression-style feature into a clear classification target, allowing algorithms like Naive Bayes and Decision Trees to predict high-value customers.

4. Exploratory Data Analysis

To understand the underlying patterns in the Online Retail dataset, we conducted an Exploratory Data Analysis (EDA) focusing on product popularity and revenue trends over time.

Key Statistics and Visualizations

Top-Selling Products: We aggregated transaction quantities by product description to identify the "best-sellers."

Observation: The top products are largely low-cost, decorative items. The "WHITE HANGING HEART T-LIGHT HOLDER" and "WORLD WAR 2 GLIDERS ASSTD DESIGNS" consistently appear as top movers, as shown in Figure 1.

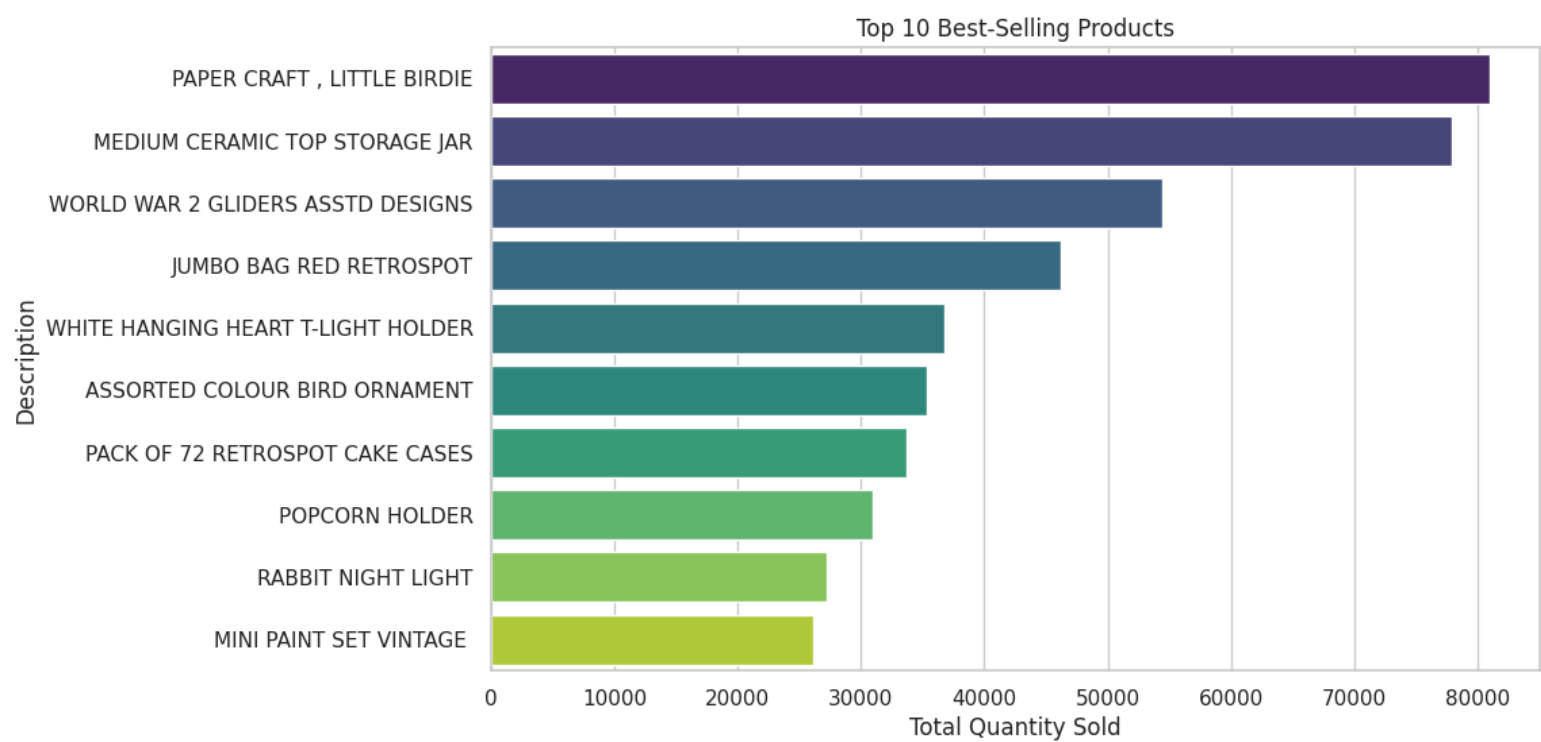


Figure 1

Monthly Revenue Trends: We calculated total monthly revenue to observe seasonal effects.

Observation: There is a distinct seasonal trend with a significant spike in revenue leading up to the end of the year (November/December), indicative of holiday shopping behavior. Conversely, the early months of the year (January/February) show slower sales, as shown in Figure 2.

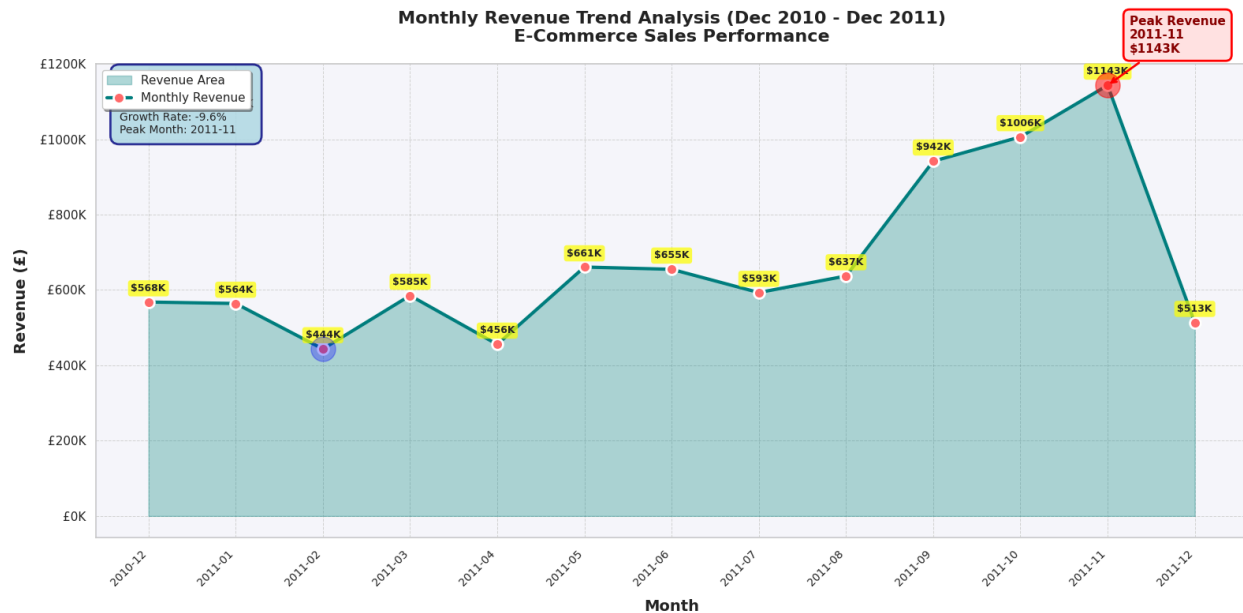


Figure 2

Trends and Outliers

- **High-Volume, Low-Price Bias:** The dataset is dominated by transactions with high quantities of low-unit-price items, suggesting a wholesaler or bulk-buyer customer base in addition to individual consumers.
- **Cancellations:** A notable number of transactions (InvoiceNo starting with 'C') were cancellations, which were removed during preprocessing to prevent skewing the revenue analysis.
- **Revenue Trend Insights:**
 - Highest Revenue: £1,143,245.77 in 2011-11
 - Lowest Revenue: £443,546.02 in 2011-02
 - Revenue Volatility (Std Dev): £217,658.48
 - Overall Growth: -9.6% from start to end

5. Association Rule Mining

To identify products that are frequently purchased together, we performed Market Basket Analysis using the Apriori algorithm from the `mlxtend` library.

Methodology

1. **Basket Creation:** Transactions were grouped by `InvoiceNo`, with products one-hot encoded (1 if present, 0 if not).
2. **Root Item Grouping:** To avoid determining rules that were too specific (e.g., "Pink Lunchbox" implies "Blue Lunchbox"), we implemented a text-processing function to group items by their **Root Product**. This allowed us to discover stronger relationships between product *types* rather than just color variants.
3. **Rule Generation:** We applied the Apriori algorithm to find frequent itemsets with a minimum support threshold and generated rules based on a minimum lift metric.

Extracted Rules and Interpretation

Interpretation of Top Association Rules:

Rule 1: Skull Paper Plates - Skull Paper Cups (Lift: 13.82)

When customers buy skull-themed plates, there's a 91% chance they also buy matching cups. This suggests themed party package opportunities

Rule 2: Children's Cutlery Spaceboy - Dolly Girl (Lift 12.57)

Strong cross-sell opportunity for children's products. This indicates that customers are buying for multiple children or for gift sets.

Business Recommendations:

- Create themed bundle packages at a discount (10-15% off)
- Optimize shelf placement for complementary items
- Implement "Frequently Bought Together" recommendations on website
- Design marketing campaigns around complete party/gift sets

6. Classification Models

We built predictive models to classify customers as "High Value" or "Low Value" to assist in targeted marketing efforts.

Feature Engineering

- Target Variable (**IsHighValue**): We performed Target Label Encoding by calculating the Customer Lifetime Value (CLV) and setting a threshold (e.g., 75th percentile). Customers above this threshold were labeled **1** (High Value), and others **0**.
- Predictors: Features included **Frequency** (number of orders), **Recency** (days since last purchase), and **Monetary** (average basket size).

Algorithms Used

1. Naive Bayes (Gaussian): A probabilistic classifier based on Bayes' theorem, assuming independence between features.
2. K-Nearest Neighbors (KNN): A non-parametric method that classifies customers based on the majority class of their nearest neighbors in the feature space.
3. Decision Tree: A tree-structured model used for its interpretability in mapping decision paths.

Evaluation & Comparison

We evaluated the models using Accuracy, Precision, Recall, and F1-Score. Contrary to initial expectations, the probabilistic approach outperformed the tree-based and distance-based methods.

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.8793	0.8768	0.8793	0.8732
KNN	0.8601	0.8550	0.8601	0.8536
Decision Tree	0.8540	0.8482	0.8540	0.8471

Classification Model Performance Analysis

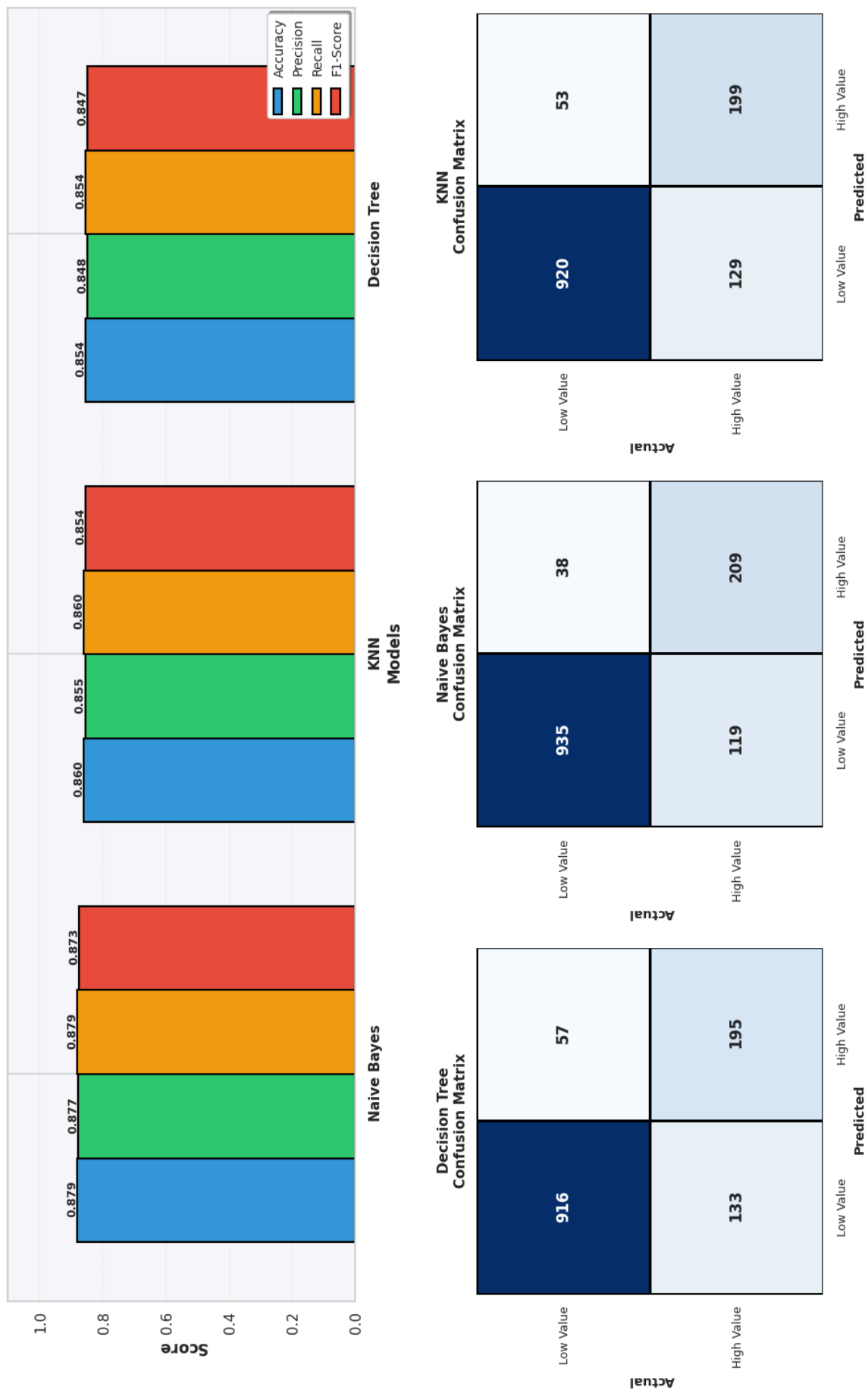


Figure 3

Key Findings

Best Performer: Naive Bayes achieved the highest accuracy (**87.93%**) and F1-Score. This suggests that the distribution of our engineered features (Recency, Frequency, Monetary) aligns well with the Gaussian assumptions of the model.

Model Suitability: While Decision Trees offer clear "if-then" rules, they slightly underperformed (85.40%), possibly due to overfitting on the complex boundaries of customer behavior.

Recommendation: For identifying high-value customers in real-time, the **Naive Bayes** model is recommended due to its superior predictive accuracy and computational efficiency.

7. Clustering and Customer Segmentation

To group customers with similar purchasing behaviors, we utilized Unsupervised Learning techniques.

Methodology (RFM Analysis)

We engineered three key features for every customer:

- **Recency:** How recently did they buy?
- **Frequency:** How often do they buy?
- **Monetary:** How much do they spend?

Clustering Algorithm

- **K-Means:** We applied the Elbow Method to determine the optimal number of clusters (k).
- **Normalization:** Due to data skewness (Pareto distribution), we applied a **Log Transformation** and **StandardScaler** to normalize the RFM values before clustering.

Customer Segments Identified

1. **The "Whales" (High Value):** High frequency, high monetary value, recent purchases.
2. **The "Loyalists":** Moderate spenders who buy frequently.
3. **The "At-Risk":** High historical spending but low recency (haven't bought in a while).
4. **The "Casuals":** Low frequency, low spending.

2D Customer Segmentation Analysis - Pairwise Feature Combinations

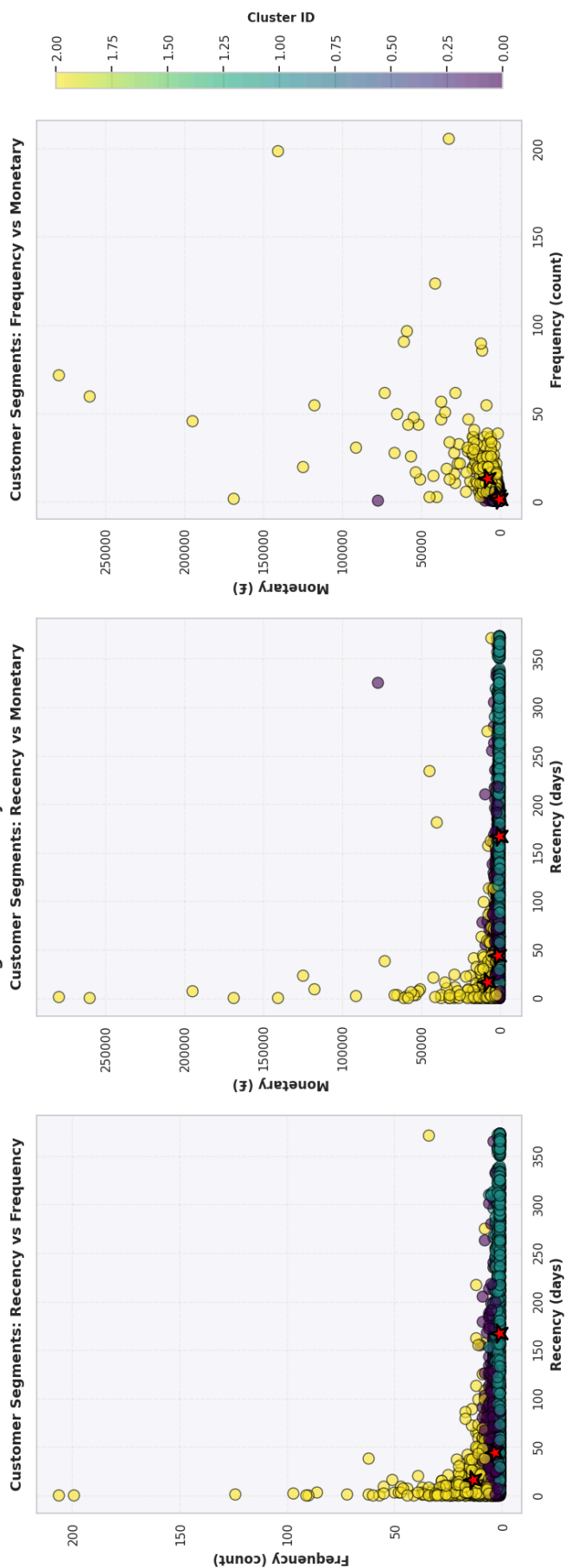


Figure 4

2D Customer Segmentation - Outlier-Filtered View for Better Cluster Separation

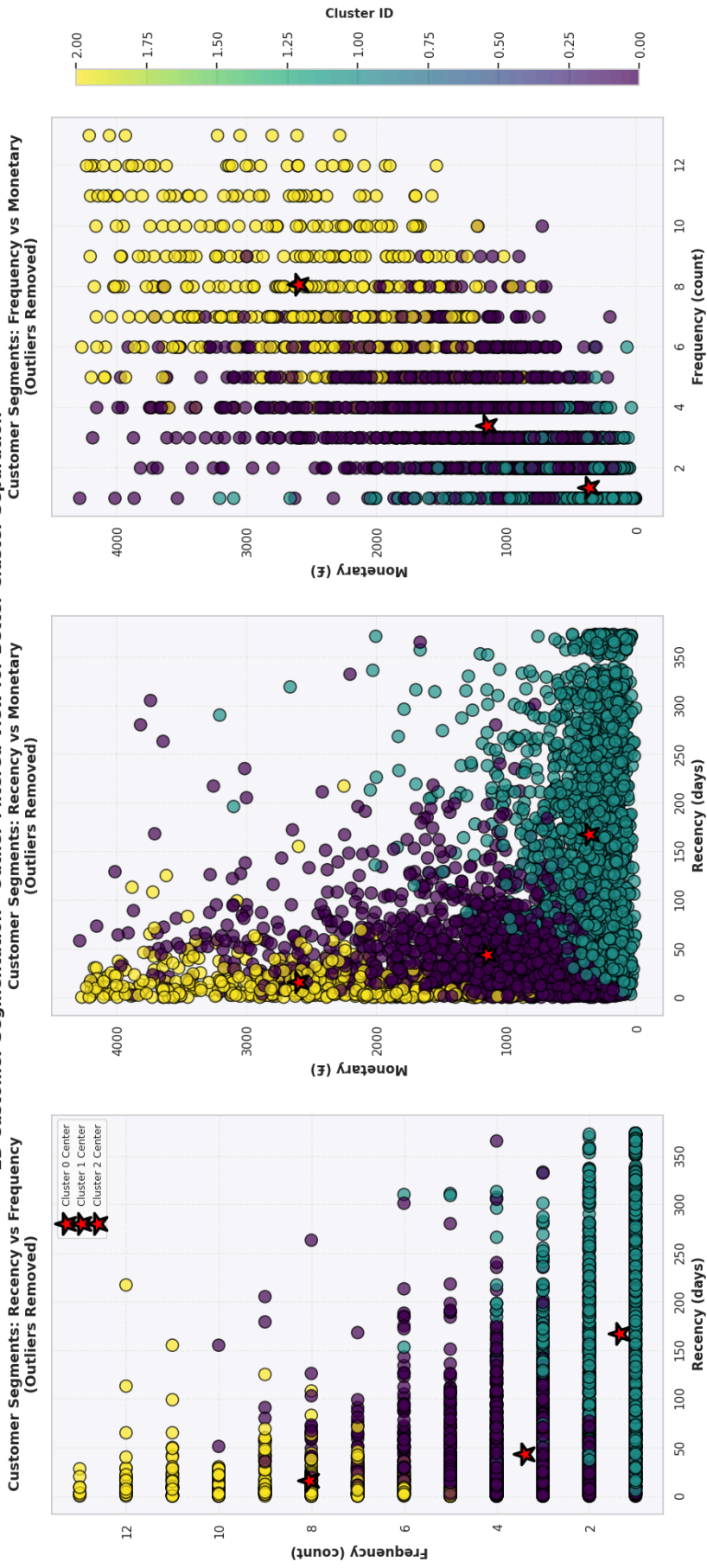


Figure 5

Figure 4 shows the clusters in a 2D representation, with 3 plots, **Recency vs Frequency**, **Recency vs Monetary** and **Frequency vs Monetary**. As can be seen, the outliers in the data cause a skew in the scale of the data, and so **Figure 5** is a scaled, filtered representation of the clusters and data points with the same plot. We opted to keep outliers, as even though the data might present them as such, they are the **High Value** customers we are trying to target. Removing them removes the information available for that segment.

8. Insights and Recommendations

Based on our data mining analysis, we propose the following business actions:

Early Detection of VIPs

Insight: The Naive Bayes classification model demonstrated ~88% accuracy in identifying "High Value" customers based on their early behavioral metrics.

Action: Implement this model into the CRM system to auto-flag new customers who fit the "High Value" probability profile, targeting them immediately with loyalty program invitations.

Inventory Bundling

Insight: Association rules revealed strong links between "Root Products" (e.g., distinct colors of the same item).

Action: Create "Complete the Set" bundles on the website. If a user adds a "Green Lunch Box" to their cart, recommend the matching "Green Water Bottle" or alternative colors of the lunch box.

Re-engagement Campaigns

Insight: The Clustering analysis identified a specific segment of "At-Risk" customers (formerly high spenders who stopped buying).

Action: Launch a specific email marketing campaign offering "We Miss You" discounts to this specific cluster.

9. Challenges and Reflections

Throughout the project, we encountered and solved several key challenges:

- **Data Skewness:**
 - *Problem:* The raw financial data followed a Pareto distribution (80/20 rule), which initially caused K-Means to fail (producing highly unbalanced clusters).
 - *Solution:* We solved this by applying a **Log Transformation** to normalize the distributions before clustering.
- **The "Variant" Trap:**
 - *Problem:* Running Association Rules on raw descriptions diluted our results because products had too many slight variations (e.g., separate entries for every color of a lunch box).
 - *Solution:* We implemented a text-processing function to group items by their **Root Product** (e.g., merging all "Lunch Box" colors), which revealed much stronger, actionable rules.
- **Data Quality:**
 - *Problem:* A significant portion (~25%) of the dataset lacked **CustomerID**.
 - *Solution:* We performed a surgical cleanup to isolate valid user behavior for segmentation tasks while preserving the integrity of the general sales analysis (EDA).

10. Tools Used

Category	Library/Tool	Usage
Data Manipulation	Pandas, NumPy	Data cleaning, aggregation, feature engineering
Visualization	Matplotlib, Seaborn	EDA plots, cluster visualization
Machine Learning	Scikit-Learn (sklearn)	K-Means, Naive Bayes, Decision Tree, KNN, Scaling
Association Rules	Mlxtend	Apriori algorithm, Association rules generation
Network Analysis	NetworkX	Visualizing product relationships
Environment	Jupyter Notebook	Development and analysis environment

11. Team Contribution Table

Member Name	Student ID	Tasks Assigned	Contribution %
Arsha Imran	2023131	Data Preprocessing, Report Writing, EDA	25%
Azeem	2023150	Association Rule Mining, Feature Engineering	25%
Hamza Bin Aamir	2023219	Classification Models, Model Evaluation	25%
Syed Areeb Zaheer	2023672	Clustering/Segmentation, Insights & Conclusion	25%